

MECH 464/563 ELEC442/EECE589
Homework Assignment #0: Linear/Matrix Algebra Refresher

Dominic Klukas
Student #: 64348378

January 12, 2026

Notes

All work shown clearly. MATLAB checks included where requested.

1 Exercise #1

Find, by hand, the eigenvalues and eigenvectors of A as well as e^{At} for

$$A = \begin{bmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Compute these in MATLAB for a few values of ω .

Solution.

We compute:

$$\det \begin{pmatrix} -\lambda & -\omega & 0 \\ \omega & -\lambda & 0 \\ 0 & 0 & -\lambda \end{pmatrix} = \lambda(\lambda^2 + \omega^2).$$

The eigenvalues satisfy $\lambda(\lambda^2 + \omega^2) = 0$, so that $\lambda = \pm i\omega, 0$. We then compute the vectors for each of these eigenvalues. We know that each eigenspace has dimension 1, since there are 3 distinct eigenvalues in a vector space of dimension 3. Thus, since $(A - 0I) \cdot (0, 0, 1) = 0$, $v_3 = (0, 0, 1)$ corresponds to $\lambda_3 = 0$. Next, we simply compute:

$$(A - \lambda_{1,2})v = \begin{pmatrix} \pm i\omega & -\omega & 0 \\ \omega & \pm i\omega & 0 \\ 0 & 0 & -\pm i\omega \end{pmatrix} v = 0$$

when $v = (\mp i, 1, 0)$ respectively, so we must have that these are the remaining two eigenvalues v_1 and v_2 .

For e^{At} , we compute, by observing the patterns that the series take and recognizing the form:

$$\begin{aligned}
e^{At} &= I + At + \frac{A^2 t^2}{2!} \dots \\
&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -\omega t & 0 \\ \omega t & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} -\omega^2 t^2/2 & 0 & 0 \\ 0 & -\omega^2 t^2/2 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \omega^3 t^3/3! & 0 \\ \omega^3 t^3/3! & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \dots \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} \sum_{n=0}^{\infty} \frac{(-1)^n \omega^{2n} t^{2n}}{(2n)!} & -\sum_{n=0}^{\infty} \frac{(-1)^n \omega^{2n+1} t^{2n}}{(2n+1)!} & 0 \\ \sum_{n=0}^{\infty} \frac{(-1)^n \omega^{2n+1} t^{2n}}{(2n+1)!} & \sum_{n=0}^{\infty} \frac{(-1)^n \omega^{2n} t^{2n}}{(2n)!} & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}.
\end{aligned}$$

We then compute $\det(e^{At} - \lambda I) = (1 - \lambda)(\lambda^2 - 2\lambda \cos(\omega t) + 1)$. The zeros are $e^{\pm i\omega t}$, and 1. Again, since we know there exist 3 linearly independent eigenvectors, we can simply plug in $(0, 0, 1)$, $(\pm i, 1, 0)$ from the last part to observe that these are indeed the corresponding eigenvectors to 1, $e^{\pm i\omega}$ respectively.

% Example MATLAB snippet:

```

omega = 2;
A = [0 -omega 0; omega 0 0; 0 0 0];
eig(A)
expm(A*t)

```

2 Exercise #2

(b) Commutation and transpose identities

Show that for any square matrix:

$$A \text{ and } e^{At} \text{ commute, } Ae^{At} = e^{At}A,$$

and

$$[e^{At}]^T = e^{A^T t}.$$

Solution.

We compute:

$$Ae^{At} = A \left(\sum_{n=0}^{\infty} \frac{A^n t^n}{n!} \right) = \sum_{n=0}^{\infty} \frac{A^{n+1} t^n}{n!} = \left(\sum_{n=0}^{\infty} \frac{A^n t^n}{n!} \right) A = e^{At} A.$$

Similarly, using the properties of the transpose (namely, that the sum of the transpose is the transpose of the sums, and the transpose multiplied by a scalar is the transpose of a map multiplied by a scalar, and transpose of a composition is composition of transposes in reversed order):

$$[e^{At}]^T = \left[\sum_{n=0}^{\infty} \frac{A^n t^n}{n!} \right]^T = \sum_{n=0}^{\infty} \frac{(A^T)^n t^n}{n!} = e^{A^T t},$$

as desired.

Finally, we have that:

$$e^{CAC^T} = \sum_{n=0}^{\infty} \frac{(CAC^T)^n}{n!} = \sum_{n=0}^{\infty} \frac{C(AC^T C)^{n-1} AC^T}{n!} = \sum_{n=0}^{\infty} \frac{CA^n C^T}{n!} = C \sum_{n=0}^{\infty} \frac{A^n}{n!} C^T = C e^A C^T.$$

I brushed by the $n = 0$ case a bit sloppily, but it follows since $I = CC^T$.

3 Exercise #3

Show that for any rotation matrix Q and any $s \in \mathbb{R}^3$,

$$(Qs)_{\times} = Q s_{\times} Q^T,$$

equivalently $(Qs) \times (Qt) = Q(s \times t)$ for all $s, t \in \mathbb{R}^3$. Hint: $a^T(s \times t) = \det[a \ s \ t]$.

Solution. We follow the hint, computing:

$$a^T(s \times t) = \det[a \ s \ t] = 1 \cdot \det[a \ s \ t] = \det Q \cdot \det[a \ s \ t] = \det[Qa \ Qs \ Qt] = a^T Q^T (Qs \times Qt).$$

Since this must be the case for any vector a , we can choose our favourite basis vector to show that the operators $(s \times t) = Q^T(Qs \times Qt)$ are equal. However, since $Q^T = Q^{-1}$ because Q is a rotation matrix (and is unitary), by applying Q to both sides of the equation we have $Q(s \times t) = (Qs \times Qt)$. That is, $Q(s_{\times}) = ((Qs)_{\times})Q$. Multiplying both sides from the left by Q^T we get

$$Q(s_{\times})Q^T = (Qs)_{\times},$$

as desired.

4 Exercise #4

Let $s \in \mathbb{R}^3$, $s = [a \ b \ c]^T$, with $s^T s = 1$. Then

$$e^{\theta s_{\times}} = I + \sin \theta (s_{\times}) + (1 - \cos \theta) (s_{\times})^2,$$

where

$$s_{\times} = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}.$$

Hint: $(s_{\times})^2 = ss^T - I$ and $(s_{\times})^3 = -s_{\times}$. Compute the eigenvalues of $e^{\theta s_{\times}}$. Give an intuitive geometrical interpretation of $e^{\theta s_{\times}}$ as an operator.

We compute, from the definition, and given the property that $(s_{\times})^3 = -s_{\times}$,

$$\begin{aligned} e^{\theta s_{\times}} &= I + s_{\times} \theta + \theta^2 (s_{\times})^2 / 2 - \frac{\theta^3 s_{\times}}{3!} + \dots \\ &= I + (s_{\times})^2 \left(-\sum_{n=0}^{\infty} \frac{\theta^{2n} (-1)^n}{(2n)!} + 1 \right) + (s_{\times}) \left(\sum_{n=0}^{\infty} \frac{\theta^{2n+1} (-1)^n}{(2n+1)!} \right) \\ &= I + (s_{\times})^2 (\cos(\theta) - 1) + (s_{\times}) \sin \theta. \end{aligned}$$

```

% Example MATLAB snippet:
s = [1; 0; 0]; s = s/norm(s);
theta = pi/3;

sx = [ 0   -s(3)  s(2);
      s(3)   0   -s(1);
      -s(2)  s(1)   0  ];

C = expm(theta*sx);

% Orthonormal check:
C'*C

% Exercise 2(c) check:
A = randn(3,3);
lhs = expm(C*A*C');
rhs = C*expm(A)*C';
norm(lhs-rhs)

% Exercise 3 check:
t = randn(3,1);
lhs = (C*s); % etc, fill in

```